2. Open upward : $x^2 = 4ay[a > 0]$

If $y < 0$, then x becomes imaginary. i.e., the curve exist for $y \geq 0$.

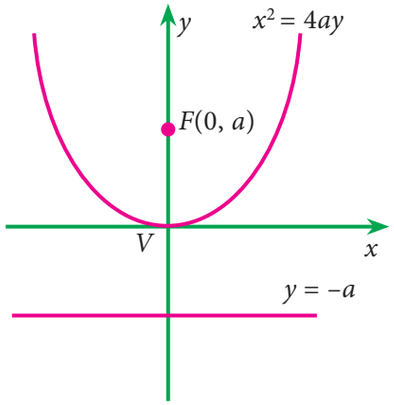


Fig. 3.12

3. Open downward : $x^2 = -4ay[a > 0]$

If $y > 0$, then x becomes imaginary. i.e., the curve exist for $y \leq 0$.

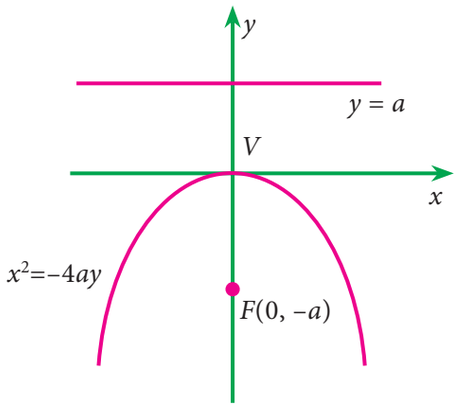


Fig. 3.13

Equations	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Axis	$y = 0$	$y = 0$	$x = 0$	$x = 0$
Vertex	$V(0, 0)$	$V(0, 0)$	$V(0, 0)$	$V(0, 0)$
Focus	$F(a, 0)$	$F(-a, 0)$	$F(0, a)$	$F(0, -a)$
Equation of directrix	$x = -a$	$x = a$	$y = -a$	$y = a$
Length of Latus rectum	$4a$	$4a$	$4a$	$4a$
Equation of Latus rectum	$x = a$	$x = -a$	$y = a$	$y = -a$

The process of shifting the origin or translation of axes.

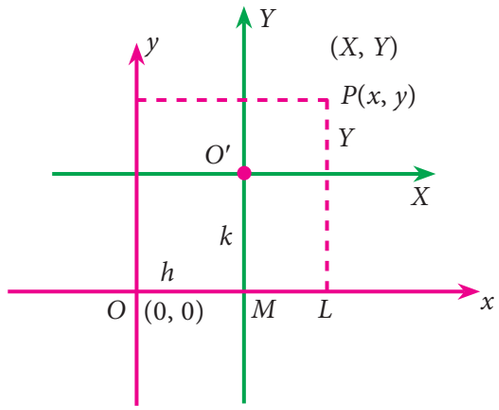


Fig. 3.14

Consider the xoy system. Draw a line parallel to x -axis (say X axis) and draw a line parallel to y – axis (say Y axis). Let $P(x,y)$ be a point with respect to xoy system and $P(X, Y)$ be the same point with respect to $XO'Y$ system.

Let the co-ordinates of O' with respect to xoy system be (h, k)

The co-ordinate of P with respect to xoy system:

$$OL = OM + ML$$
$$= h + X$$
$$\text{i.e) } x = X + h$$
$$\text{similarly } \quad y = Y + k$$

$\therefore$ The new co-ordinates of P with respect to $XO'Y$ system.

$$X = x - h \text{ and } Y = y - k$$

3.5.4 General form of the standard equation of a parabola, which is open rightward (i.e., the vertex other than origin):

Consider a parabola with vertex V whose co-ordinates with respect to XOY system is $(0, 0)$ and with respect to xoy system is (h, k). Since it is open rightward,

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